

Hashing Concepts & Hash Functions Data Structures

> **Definition:** Hashing is a technique that transforms a search key K into a table index $h(K)$ using a **Hash Function**, enabling average constant time $O(1)$ insertion, deletion, and lookup operations in a **Hash Table**.

1. Detailed Technical Explanation

1. The Load Factor (ϕ)

The load factor represents the density of items stored in a hash table of size m with n keys:

2. Common Hash Functions

1. Division Method (Modulo Arithmetic)

$h(k) =$

2. Mid-Square Method

1. Square the key: k^2 .

2. Extr

3. Folding Method

Divide the key digits into equal parts of size r , and sum them together (ignoring overflow carry).

$- k^r$

4. Multiplication Method

$h(k) =$

3. Properties of a Good Hash Function

1. **Uniform Distribution:** Distributes keys evenly across all table slots to minimize collisions.

2. **De**

4. Quick Recall Flow

Key ->